

Blockchain based Anonymous Authentication Framework for Healthcare IoT

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Abstract—Healthcare Internet of Things (IoT) connects patients, medical devices, and healthcare providers to enable real-time monitoring and efficient healthcare services. However, these systems face critical challenges in secure authentication, as sensitive medical data is highly vulnerable to forgery, malicious key generation, and privacy breaches. Ensuring anonymity is equally essential to prevent identity leakage, which may compromise patient confidentiality and trust. Existing certificateless signature (CLS) schemes provide a foundation for lightweight authentication with anonymity, integrity, and resistance to replay. Nevertheless, these schemes remain limited in handling malicious key generation and limited flexibility in dynamic healthcare environments. To address these issues, this work proposes an enhanced certificateless signature (CLS) scheme that achieves anonymity while resisting malicious key generation, secret key exposure, and signature forgery. The scheme ensures data integrity through a hybrid framework that supports both blockchain-based and non-blockchain deployment, providing flexibility for different healthcare environments. In addition, the proposed approach can be extended with revocation mechanisms to eliminate malicious users and attackers, thereby enhancing long-term system resilience.

Index Terms—Healthcare Internet of Things (IoT), certificateless signature (CLS), authentication, anonymity, data integrity, blockchain, security, revocation.

I. INTRODUCTION

The swift expansion of IoT into the realm of healthcare has sparked a revolutionary shift in how healthcare services are provided and received. The Healthcare Industrial Internet of Things (Healthcare IIoT) revolutionizes traditional care by virtue of its capabilities to perceive everything, analyze intelligently, and operate independently of human interference [1], [2]. In healthcare IIoT, researchers attach tiny biosensors, including pulse oximeters, ECG patches, continuous glucose monitors, and blood pressure cuffs, directly onto patient's body to sense the patient's physiological parameters, which include the level of blood oxygen saturation, heart rate, respiration rate, and body temperature [3]. Due to the huge volume of data generated by these sensors, as well as the limited processing capabilities of the wearable devices themselves,

medical researchers must outsource EHR data to cloud-based and distributed ledgers for persistent storing and analyzing, not to mention inter-hospital EHR data exchange [4], [5]. In general, the open and heterogeneous environment of wireless communication in Healthcare IIoT subjects the EHR data to multiple kinds of attacks. Some attacks that adversaries could launch against the EHR data include forgery attacks, man-in-the-middle attacks, replay attacks, and eavesdropping attacks [6]. The disclosure and unauthorized modification of sensitive health information not only damages patient privacy but can directly threaten patient safety by leading to erroneous clinical decisions—incorrect drug dosages, missed diagnoses, or falsified laboratory results—thereby deteriorating the integrity of the doctor-patient relationship [7]. Moreover, in smart healthcare systems, both medical personnel and patients frequently access the system to retrieve or update health-related information; unauthorized access by adversaries may lead to catastrophic consequences including identity theft, data leakage [8], and the misuse of personal health data. Therefore, ensuring the authenticity, integrity, confidentiality, and availability of EHR data, while simultaneously establishing a secure and lightweight authentication mechanism, has become a critical and urgent research imperative.

Digital signatures serve as a fundamental cryptographic tool to provide identity authentication, data integrity verification, and non-repudiation in electronic data transmission [9]. In the context of Healthcare IIoT, a robust digital signature framework must satisfy several stringent yet often conflicting requirements: (i) computational efficiency suitable for resource-constrained wearable devices with limited battery life and processing power; (ii) minimal communication overhead to accommodate bandwidth-limited sensor networks; (iii) strong privacy guarantees to comply with regulatory frameworks such as HIPAA and GDPR; and (iv) mutual authentication between communicating parties to prevent impersonation and unauthorized EHR access [10], [11]. Critically, we observe that not all healthcare interactions demand the same level of infrastruc-

ture. A routine telemedicine consultation between a patient and a doctor requires fast, lightweight mutual authentication—but does not necessarily warrant the overhead of a full blockchain transaction. Conversely, uploading an official EHR for cross-hospital sharing or regulatory auditing mandates immutable on-chain storage with verifiable provenance. This fundamental asymmetry in operational requirements motivates the hybrid architecture proposed in this work.

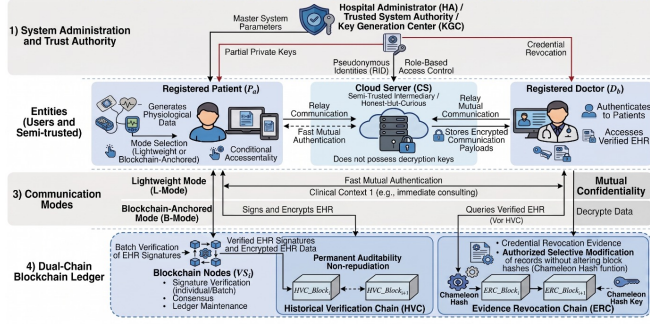


Fig. 1. Sample figure caption

A. Motivation

The construction of an efficient and reliable authentication scheme for Healthcare IIoT requires rigorous attention to the cryptographic foundation as well as the system deployment architecture. The traditional PKI-based signature mechanisms face unaffordable certificate management complexity when scaling up the number of involved parties to cover patients, doctors, nurses, and sensor nodes. In addition, the IBC approach dispenses with certificates but suffers from the key escrow issue. The key generation center (KGC) holds all the secret keys, which enables any adversary controlling the KGC to forge signatures of arbitrary users. Such an attack is completely intolerable in clinical practice [12]. On the other hand, CL-PKC solves both issues by distributing the private key among the KGC and the user. However, most existing CLS approaches do not satisfy Girault’s Level-3 security requirements [13].

Apart from the cryptographic foundation, real-world deployment on energy-limited clinical wearables entails strict resource limitations. The most costly elliptic curve cryptographic primitive is the scalar multiplication (T_{Mu}). Most provably secure CLS frameworks demand at least one scalar multiplication for generating a digital signature. Such a requirement consumes significant battery power and creates unacceptable latency, making it impractical for real-time EHR transfer applications like ECG patches and pulse oximeters [14]. Furthermore, current CLS schemes with formal security proofs mandate four or more distinct cryptographic hash function invocations per signature, compounding the computational burden on constrained processors [15].

Authentication itself also faces many obstacles. Currently, a majority of healthcare systems adopt biometric fuzzy extractors that generate cryptographic keys from noisy bio-

metric signals [16]. Although biometric fuzzy extraction offers an intriguing prospect, the scheme involves non-trivial False Rejection Rates (FRR) when applied to physiological stress factors common in clinical scenarios such as trembling hands, sweating, and skin abrasions, resulting in legitimate users being denied access in emergency situations. Hence, a credential-based multi-factor authentication solution becomes indispensable. In addition, most CLS-based authentication protocols offer a one-size-fits-all authentication mechanism without regard for user identity at the cryptographic level [17]. In contrast, patients and doctors in healthcare settings have completely different roles.

Most importantly, the current blockchain-based healthcare authentication systems exhibit inflexibility in terms of architectural design, requiring either full blockchain overhead for each transaction or no blockchain at all, which compromises accountability. Yet, there exists no patient-selectable hybrid architecture, enabling participants to decide whether off-chain authentication and/or EHR storage using blockchain technology is necessary.

B. Our Contributions

In order to overcome the above-mentioned obstacles, we introduce a *Hybrid Blockchain-Based Certificateless Conditional Anonymous Authentication Framework for Secure EHR Sharing in Healthcare IIoT*. This framework is distinctively structured on a dual-mode architecture that enables participants to select the appropriate security-performance trade-off for each clinical interaction. The principal contributions of this work are summarized as follows.

- **Zero- T_{Mu} Certificateless Signature Scheme:** We propose a pairing-free CLS scheme with no elliptic curve scalar multiplication operations in the online signing stage by leveraging pre-computed Secret Identifiers. The proposed CLA-based hash function significantly decreases the number of hashing operations to two, resulting in a very lightweight signature generation cost of $0T_{Mu} + 2T_{ha} + 6T_{mu}$.
- **Dual-Mode Hybrid Architecture:** We construct a user-selective approach that allows two different modes of operation: a Lightweight Mode for peer-to-peer off-chain mutual authentication during telemedicine consultations, and a Blockchain-Anchored Mode for immutable on-chain EHR entries and cross-hospital EHR exchange.
- **Dual-Chain Blockchain with Chameleon Revocation:** For the on-chain mode, we implement a dual-chain architecture consisting of a Historical Verification Chain (HVC) chain for verifying EHR entries and an Evidence Revocation Chain (ERC) that employs Chameleon Hashing technique to selectively revoke EHR credentials without breaking the blockchain structure.
- **Role-Aware Multifactor Mutual Authentication:** We incorporate a deterministic multifactor verification mechanism with knowledge and possession tokens in lieu of biometric fuzzy extractors. The model inherently includes the roles of the actors (PATIENT/DOCTOR) within

the pseudonyms to prevent cross-role impersonation of EHRs.

- **Performance Analysis:** Thorough theoretical cost analysis and simulation on a testbed show the superior latency, throughput, and communication overhead of our authentication framework relative to current authentication schemes for health care.

The remainder of this paper is structured as follows: Section II reviews related literature. Section III outlines the mathematical preliminaries. Section IV describes the system and threat models. Section V details the proposed certificateless signature and dual-mode blockchain architectures. Section VI provides the formal security proofs, while Section VII details the mutual authentication protocol. Section VIII evaluates performance, followed by concluding remarks in Section IX.

II. RELATED WORKS

This section critically reviews the relevant literature along two converging research avenues: certificateless signature schemes for IoT systems and blockchain-enabled authentication methods for medical data exchange. We then provide an analysis that contrasts our hybrid approach with existing literature.

A. Certificateless Signature Schemes for IoT

Certificateless public key cryptography (CL-PKC), proposed by Al-Riyami and Paterson [18], addresses the disadvantages of traditional PKI, which require costly certificate management, and those of IBC, which suffer from a key escrow problem. The full private key of each user is generated based on the KGC-issued partial private key and a personal secret chosen by the user, guaranteeing that no forgery can be carried out by the KGC alone. Since CL-PKC was first proposed, many CLS schemes have been put forward, with a clear evolutionary trend from pairing-based to pairing-free constructions.

Some early pairing-based CLS schemes [19], [20] attained their security guarantee under the Bilinear Diffie-Hellman assumption, but the computation cost associated with their bilinear pairings became excessive. Hence, the idea of achieving CLS without using bilinear pairings through ECC proved to be more attractive. Yet, devising such a pairing-free CLS scheme is extremely difficult and leads to an endless loop of cryptanalysis and improvement.

One of the first attempts towards constructing a pairing-free CLS scheme for IIoT applications is made by Karati et al. [20], which claimed to provide existential unforgeability. Soon afterward, another pairing-free CLS scheme was developed by Jia et al. [21] to claim its security. Later on, Du et al. [23] and Xiang et al. [24] pointed out that Jia et al.'s scheme is susceptible to Type-I (external) adversary forgery attacks, and hence, devised better schemes accordingly. In turn, Ma et al. [22] successfully launched forgery attacks on Du et al.'s and Xiang et al.'s schemes and showed their insecurity against forgery.

Thumbur et al. [11] introduced an efficient CLS scheme specifically for resource-constrained environments. Xu et al. [21] subsequently proved that Thumbur et al.'s scheme could be trivially forged by a Type-I adversary. Wang et al. [1] then demonstrated that Xu et al.'s own repair was still vulnerable to replay attacks and KGC compromise, and proposed a Blockchain-Based Certificateless Conditional Anonymous Authentication (BCCA) scheme with precomputed key materials and batch verification. However, Shim [23] later exposed a critical flaw in Wang et al.'s signing algorithm: the omission of a per-session random nonce allowed an attacker to recover a user's complete private key from merely two intercepted signatures.

Addressing the persistent insecurity of pairing-free CLS schemes against malicious KGCs, Qiao et al. [24] formally demonstrated that virtually all preceding pairing-free CLS constructions fail to achieve Girault Level-3 security—the highest trust level, wherein even a fully compromised KGC cannot forge valid signatures. Qiao et al. proposed a provably secure CLS scheme with anonymity that achieves Level-3 under the hardness of the discrete logarithm problem (DLP) in the random oracle model. Despite its strong security guarantees, the scheme requires one online elliptic curve scalar multiplication (T_{Mu}) and four distinct hash invocations per signature—overhead that remains prohibitive for battery-powered clinical sensors transmitting continuous physiological data streams.

B. Blockchain-Assisted Healthcare Authentication

The healthcare ecosystem has widely implemented blockchain technology to offer immutable audit trails and ensure trust between multiple parties by enabling data provenance. Tang et al. [25] provided a robust authentication system based on blockchains and smart contracts for EHR applications, although they didn't utilize certificateless cryptography and therefore carried over the drawbacks related to certificate management of PKI schemes. Another example is that of Yazdinejad et al., who designed a decentralized authentication system for distributed hospital network environments [25], even though their protocol was neither formally verified nor satisfied the condition of anonymity.

Wireless body area networks (WBANs) constitute one of the most significant parts of the Healthcare IIoT infrastructure, where various authentication schemes have been developed. For instance, He et al. [27] introduced a protocol for anonymous authentication in WBANs and proved its security properties; however, such protocol uses bilinear pairing calculations. Xiong introduced a more resource-friendly anonymous certificateless remote authentication scheme for WBANs [29]; at the same time, Xiong and Qin revealed weaknesses in large-scale deployment of this protocol. Next, Liu et al. [26] enhanced existing two-layer authentication protocol, yet still without blockchain functionality or mutual authentication support. Recently, Meher et al. [28] proposed an anonymous mutual authentication scheme for WBANs based on DLP, ECDLP and bilinear pairing operations. Meher et al. achieved

TABLE I
TRANPOSED FEATURE COMPARISON

Scheme	Healthcare	Certificateless	Provable	Conditional	Mutual	Blockchain	Revocation	Girault L3	Zero T_{Mu}	Hybrid
He [27]	✓	×	✓	✓	✓	×	×	×	×	×
Ma [22]	×	✓	✓	✓	✓	×	×	×	×	×
Wang [1]	✓	✓	×	✓	×	✓	✓	×	×	×
Qiao [24]	×	✓	✓	✓	×	×	×	✓	×	×
Tang [25]	✓	×	✓	✓	✓	✓	×	×	×	×
Meher [28]	✓	×	✓	✓	✓	×	×	×	×	×
Soni [26]	✓	×	✓	×	✓	✓	×	×	×	×
Ours	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

TABLE II
COMPARISON OF SIGNATURE/AUTHENTICATION SCHEMES: HASH CALLS, PRECOMPUTATION, AND SECURITY

Paper	Type	Precomp. Keys	Hash Calls		Security Scheme
			Sign	Verify	
Zhu et al. [2]	CLS	No	3	3	CDH
Tsai et al. [19]	CLS (Pairing)	No	3	4	BDH
Karati et al. [20]	CLS	No	3	3	DLP
Jia et al. [21]	CLS	No	3	3	CDH
Thumbur et al. [11]	CLS	No	2	3	ECDLP
Ma et al. [22]	CLS	No	4	4	CDH
Qiao et al. [24]	CLS	No	4	4	DLP
Zhou et al. [34]	CLS (Agg.)	No	3	3	CDH
Yang et al. [31]	CLS	No	3	3	ECDLP
Bansal et al. [32]	CLS	No	3	3	ECDLP
He et al. [27]	Auth	No	2	3	BDH
Meher et al. [28]	Auth	No	3	3	DBDH
Xiong et al. [29]	CLS-Auth	No	3	4	ECDLP
Tang et al. [25]	BC-Auth	No	2	2	CDH
Soni et al. [26]	BC-Auth	No	3	3	CDH
Wang et al. [1]	BCCA	Yes	4	3	ECDLP
Ours (CLS)	CLS	Yes	2	2	ECDLP
Ours (BCCA)	BCCA	Yes	3	3	ECDLP

CLS = Certificateless Signature; BCCA = Blockchain-Based Certificateless Conditional Anonymous Authentication; BC-Auth = Blockchain Authentication; Agg. = Aggregate; GL3 = Girault Level-3;; Std. = Standard Model; BDH = Bilinear DH; CDH = Computational DH; ECDLP = Elliptic Curve DLP; DBDH = Decisional BDH.

the strong security guarantees and mutual authentication, but's hybrid approach inherently retains the computational cost of pairings, limiting its applicability on ultra-constrained clinical sensors.

The research work by Soni et al. [26] proposed Blockchain-Based User Authentication and Data-Sharing (BC-UADS) framework for healthcare industries that used consortium blockchain model and the Proof-of-Reputation consensus mechanism. The framework offers support for mutual authentication and multi-hospital EHR sharing but did not use certificateless cryptography and, therefore, cannot withstand certificate attack. Zhu et al. [2] created certificateless redactable signature scheme with batch verification for Healthcare IoT and integrated the concept of privacy-preserving data sharing and efficient verification of multi-signature. This scheme does not offer conditional anonymity and role-based access control.

The most important architectural issue inherent almost in all the present-day blockchain healthcare frameworks is their rigidity and single-mode approach; namely, the traffic of authentication is inevitably passed through the blockchain regardless of the particular case. Wang et al. [1] provided a partial solution to the problem of flexible revocation through introduction of a dual chain system that allowed separating historical records for verification and evidence about revocation with the help of chameleon hashing. Unfortunately, their blockchain still requires passing everything through on the blockchain without providing any flexibility.

C. Literature Summary and Comparison

To synthesize the current state of research across both certificateless cryptography and blockchain-based healthcare authentication, Table ?? presents a feature-level comparison

of the strongest representative schemes against our proposed framework. By selecting highly competitive recent protocols—many of which achieve provable security, mutual authentication, and conditional anonymity—the comparison reveals that while individual works excel in specific domains, no existing scheme simultaneously bridges all gaps. Our proposed framework is unique in aggregating pairing-free certificateless design, Girault Level-3 security, zero online scalar multiplications, and an auditable blockchain-based revocation system into a unified, patient-selectable hybrid architecture.

Cryptographically, our scheme builds upon the Level-3 security foundation established by Qiao et al. [24] but substantially reduces the signing overhead through precomputed Secret Identifier (SID) arrays and a novel Carry Look-Ahead (CLA)-based hash derivation. This design eliminates all online scalar multiplications and reduces hash invocations from four to two per signature. Architecturally, we inherit the dual-chain chameleon hash revocation model pioneered by Wang et al. [1] but critically decouple the authentication layer from the blockchain layer. This breakthrough drops the online signing cost from $1, T_{Mu} + 4, T_{ha}$ down to $0, T_{Mu} + 2, T_{ha}$, making it significantly lighter than Qiao et al.’s model. This decoupling enables a patient-selectable hybrid operational mode: a *Lightweight Mode* for direct peer-to-peer mutual authentication during routine telemedicine sessions, and a *Blockchain-Anchored Mode* for immutable, auditable EHR storage. To the best of our knowledge, no prior work offers this dual-mode flexibility within a unified certificateless framework.

III. PRELIMINARIES

This section introduces the mathematical foundations, cryptographic definitions, and the formal security model underlying the proposed hybrid certificateless authentication framework. Table III summarizes the principal notation used throughout the remainder of this paper.

A. Elliptic Curve Cryptography

Let q be a large prime and $\mathbb{F}_q$ a finite field. A non-singular elliptic curve over $\mathbb{F}_q$ is defined as:

$$E(\mathbb{F}_q) : y^2 = x^3 + ax + b \pmod{q} \quad (1)$$

where $a, b \in \mathbb{F}_q$ and $4a^3 + 27b^2 \not\equiv 0 \pmod{q}$. The set of all points on $E(\mathbb{F}_q)$ together with the point at infinity $\mathcal{O}$ forms an additive cyclic group G with generator P and prime order q . For any point $Q \in G$ and scalar $k \in \mathbb{Z}_q^*$, the scalar multiplication $Q = kP$ is efficiently computable, but its inverse—recovering k given Q and P —is computationally intractable.

B. Hardness Assumptions

Definition 1 (Discrete Logarithm Problem — DLP). Given the additive cyclic group G of prime order q with generator P , and a random point $Q = aP$ where $a \in \mathbb{Z}_q^*$, the DLP is to compute a from (P, Q) . The DL assumption states

TABLE III
NOTATION SUMMARY

Symbol	Description
κ	Security parameter
$E(\mathbb{F}_q)$	Elliptic curve over the finite field $\mathbb{F}_q$
G	Additive cyclic group of prime order q on $E(\mathbb{F}_q)$
P	Generator of G
$\mathcal{O}$	Point at infinity (identity element of G)
s	System master secret key
$P_{pub} = sP$	System master public key
P_{pub1}, P_{pub2}	CLA-derived domain-separated public keys (blockchain mode)
RID_i	Real identity of user U_i
ID_i	Pseudonym (anonymous identity) of U_i
x_i	User’s self-chosen secret value
$upk_i = x_i P$	User’s public key
d_i (or psk_i)	Partial private key issued by KGC/HA
$gpk_i = k_i P$	Partial public key corresponding to psk_i
$SK_i = (x_i, psk_i)$	User’s full private key
$PK_i = (upk_i, gpk_i)$	User’s full public key
$SID_{i,k}$	k -th precomputed secret key from the SID set
$KID_{i,k} = SID_{i,k} \cdot P$	k -th precomputed public key from the KID set
ψ	CLA carry-in value, $\psi = x\text{-coord}(KID_k) \bmod q$
H_0, H_1, H_2, H_3	Cryptographic hash functions (modeled as random oracles)
$\oplus$	Bitwise XOR operation
T_{Mu}	Time cost of one EC scalar multiplication
T_{Add}	Time cost of one EC point addition
T_{ha}	Time cost of one general hash operation
T_{mu}	Time cost of one modular multiplication
ΔT	Maximum allowable timestamp deviation (freshness window)

that for any probabilistic polynomial-time (PPT) adversary $\mathcal{A}$, the advantage

$$\text{Adv}_{\mathcal{A}}^{DL}(\kappa) = \Pr[\mathcal{A}(P, aP) = a] \quad (2)$$

is negligible in κ .

Definition 2 (Elliptic Curve Discrete Logarithm Problem — ECDLP). Given a point $Q \in G$ and the generator P of G , the ECDLP is to find $x \in \mathbb{Z}_q^*$ such that $Q = xP$. Finding such x is computationally infeasible for appropriately chosen parameters [30].

Definition 3 (Elliptic Curve Computational Diffie-Hellman Problem — ECCDHP). Given $(P, aP, bP) \in G$ for unknown $a, b \in \mathbb{Z}_q^*$, the ECCDHP is to compute abP . The ECCDH assumption states that no PPT adversary can solve this problem with non-negligible probability.

Remark: The security of the proposed CLS scheme and mutual authentication protocol relies on the hardness of the DLP. The ECCDHP underpins the security of the Diffie–

Hellman-based session key agreement in the mutual authentication phase.

C. Girault's Security Levels for CL-PKC

Girault [13] defined a hierarchy of trust levels for key generation centers (KGCs) in certificateless and certificate-based cryptographic schemes. Hu et al. subsequently extended this classification to the CLS setting:

- **Level 1:** The KGC possesses sufficient information to compute users' full private keys and can impersonate any user.
- **Level 2:** The KGC does not know users' full private keys but can generate fraudulent partial keys that pass verification.
- **Level 3:** The KGC cannot forge valid signatures even by fabricating partial private keys, as inconsistencies with user public keys are detectable.

Significance: Level 3 provides the strongest trust guarantee: even a fully compromised Hospital Administrator (serving as the KGC) cannot forge a valid patient or doctor signature. The proposed scheme achieves Girault Level-3 by binding the partial private key psk_i to the user's public key upk_i within the hash computation

$$h_1 = H_1(ID_i, gpk_i, upk_i, P_{pub}, E_i) \quad (3)$$

ensuring that any substitution by the KGC is publicly detectable.

D. Chameleon Hash Function

A Chameleon Hash function [35], introduced by Krawczyk and Rabin, is a trapdoor collision-resistant hash function. It consists of the following algorithms:

- **Setup**(λ): Outputs public parameters pp .
- **KeyGen**(pp): Outputs (HK, ck) , where HK is the hash key and ck is the trapdoor key.
- **Hash**(HK, m, ζ): Outputs hash value CH .
- **Verify**(HK, m, ζ, CH): Verifies correctness of CH .
- **Forge**(ck, m, ζ, m'): Computes ζ' such that

$$\text{Hash}(HK, m, \zeta) = \text{Hash}(HK, m', \zeta'). \quad (4)$$

Security Properties:

- **Collision Resistance:** Without ck , finding collisions is infeasible.
- **Trapdoor Collision:** With ck , controlled collisions are efficiently computable.

In our framework, the Hospital Administrator holds the trapdoor key to enable controlled modification of revocation evidence without altering blockchain integrity.

E. Carry Look-Ahead (CLA) Adder — Algebraic Hash Derivation

The Carry Look-Ahead (CLA) adder computes carry bits in parallel:

$$C_{i+1} = G_i + P_i \cdot C_i \quad (5)$$

where $G_i = A_i \cdot B_i$ and $P_i = A_i \oplus B_i$.

It achieves this through two fundamental Boolean logic components: the "generate" signal (which produces a carry independent of previous states) and the "propagate" signal (which passes a carry forward if one exists).

In conventional pairing-free certificateless signature schemes, executing multiple independent cryptographic hash functions per transaction incurs significant computational overhead, rendering them suboptimal for continuous transmission in resource-constrained IIoT devices. We innovatively repurpose the structural logic of the CLA—specifically its generate and propagate characteristics—and map it onto the finite field $\mathbb{Z}_q^*$.

F. Forking Lemma

The Forking Lemma is used to prove signature security in the Random Oracle Model.

Lemma 1: If a PPT adversary $\mathcal{A}$ produces a valid forgery with probability ε after Q_h hash queries, then a PPT algorithm can extract two forgeries with probability:

$$\varepsilon' \geq \left(1 - \frac{1}{e}\right) \frac{\varepsilon}{Q_h}. \quad (6)$$

Application: The lemma is applied to hash oracles H_1 and H_2 to extract two valid signatures, enabling reduction to the DLP and proving unforgeability.

IV. METHODOLOGY

In this section, we present the complete construction of the proposed Hybrid Blockchain-Based Certificateless Conditional Anonymous Authentication Framework for Healthcare IIoT. The framework consists of a common construction shared by both operational modes, a Lightweight Mode for off-chain peer-to-peer authentication, and a Blockchain-Anchored Mode for on-chain EHR operations. Each mode shares the same registration, key generation, and login infrastructure, diverging only at the signature and authentication layers. The scheme supports user revocation and on-chain evidence modification.

A. System Model

System Model: The proposed dual-mode hybrid framework operates in a Healthcare IIoT environment comprising six entities (Fig. 1).

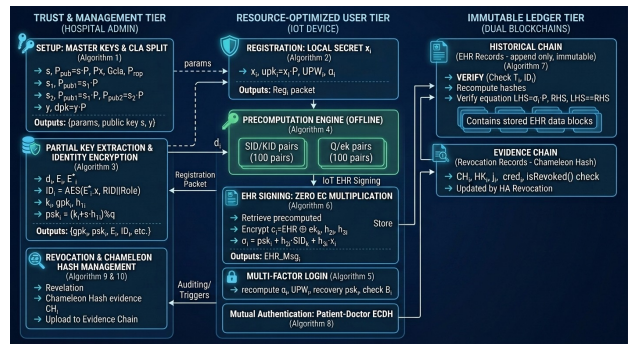


Fig. 2. Sample figure caption

Algorithm 1 Proposed Modified CLS Scheme

Entities: KGC, User U_i , Verifier V_j

Phase 1: Setup(κ)

- 1) Select group (G, q, P) ; choose $s \in \mathbb{Z}_q^*$
- 2) Compute $P_{pub} = sP$; define H_0, H_1, H_2
- 3) Publish $Params$

Phase 2: Key Generation

- 1) $x_i \in \mathbb{Z}_q^*$, $X_i = x_i P$
- 2) KGC computes:

$$R_i = rP, \quad h_1 = H_1(P_{pub}, ID_i, R_i, X_i)$$

$$d_i = r + sh_1$$
- 3) Keys: $SK_i = (x_i, d_i)$, $PK_i = (X_i, R_i)$

Phase 3: Offline Precomputation

- 1) Generate session pairs:

$$SID_k = v_k x_i + h_1, \quad KID_k = SID_k P$$

Phase 4: Signing

- 1) $h_2 = H_2(PK_i, m, Time, KID_k)$
- 2) $\psi = x(KID_k)$
- 3) $h_3 = h_1 h_2 + (h_1 \oplus h_2) \psi$
- 4) $h_4 = h_3(h_2 + (h_2 \oplus h_3))$
- 5) $\sigma = h_4 SID_k + h_3 d_i + h_2 x_i$
- 6) Output (σ, KID_k)

Phase 5: Verification

- 1) Recompute h_1, h_2, h_3, h_4
- 2) Verify:

$$\sigma P = h_4 KID_k + h_3(R_i + h_1 P_{pub}) + h_2 X_i$$

$$= 0$$

Hospital Administrator (HA): Acts as the trusted authority and KGC, generating system parameters, issuing partial keys and pseudonymous identities, assigning roles, and managing revocation via the Evidence Chain. The model also considers a compromised HA.

Patient (P_a): Equipped with wearable sensors (e.g., ECG, SpO₂, glucose monitors) generating health data. The patient selects Lightweight or Blockchain-Anchored mode, performs mutual authentication, and uploads signed and encrypted EHRs when required.

Doctor (D_b): A registered practitioner who authenticates with patients to access records. In blockchain mode, retrieves verified EHRs and decrypts them using private key y .

Cloud Server (CS): A semi-trusted relay for authentication messages. It stores encrypted data but cannot access plaintext, following an honest-but-curious model.

Blockchain Nodes (VS_i): Consortium nodes that verify certificateless signatures, maintain consensus, and support batch verification.

Dual-Chain Ledger: Consists of (i) Historical Verification Chain (HVC) for immutable EHR logs and (ii) Evidence Revocation Chain (ERC) for revocation records, where Chameleon Hashing enables controlled modifications without changing block hashes.

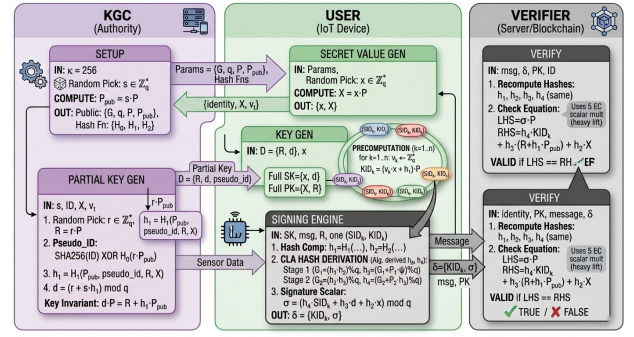


Fig. 3. Sample figure caption

V. PROPOSED FRAMEWORK

In this section, we present the complete construction of the proposed Hybrid Blockchain-Based Certificateless Conditional Anonymous Authentication Framework for Healthcare IIoT. The framework consists of a common construction shared by both operational modes, a Lightweight Mode for off-chain peer-to-peer authentication, and a Blockchain-Anchored Mode for on-chain EHR operations. Each mode shares the same registration, key generation, and login infrastructure, diverging only at the signature and authentication layers. The scheme supports user revocation and on-chain evidence modification.

A. Common Construction

This phase consists of the following polynomial-time algorithms executed during system initialization and user onboarding. Their outputs serve both operational modes.

Setup: In this phase, the Hospital Administrator (HA) generates the system public parameters based on a security parameter κ .

- 1) Given κ , HA selects an elliptic curve E over a finite field $\mathbb{F}_q$, and an additive cyclic group G of prime order q with generator P .
- 2) HA randomly selects $s \in \mathbb{Z}_q^*$ as the master secret key and computes the system master public key $P_{pub} = sP$.
- 3) HA performs CLA-based public key splitting for the Blockchain-Anchored Mode. Let $P_x = x\text{-coordinate}(P) \bmod q$. HA computes $G_{cla} = (s \cdot P_x) \bmod q$ and $Prop = (s \oplus P_x) \bmod q$, and derives $s_1 = G_{cla}$ and $s_2 = (G_{cla} + Prop) \bmod q$, yielding $P_{pub1} = s_1 \cdot P$ and $P_{pub2} = s_2 \cdot P$.
- 4) HA generates a decryption key pair: private key $y \in \mathbb{Z}_q^*$ (assigned to authorized doctors) and public key $dpk = y \cdot P$.
- 5) HA defines cryptographic hash functions modeled as random oracles: $H_0 : G \times \{0, 1\}^* \rightarrow \{0, 1\}^*$, $H_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H_2 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$, $H_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^*$.
- 6) HA publishes:

$$params_L = \{E, G, q, P, P_{pub}, H_0, H_1, H_2\}$$
 for Lightweight Mode and

$$params_B = \{E, G, q, P, P_{pub}, P_{pub1}, P_{pub2}, dpk, H_1, H_2, H_3\}$$
 for Blockchain Mode.

Algorithm 2 Proposed Blockchain-Based authentication scheme

Entities: HA, Patient P_a , Doctor D_b , CS, BC

Phase 1: Setup(κ)

- 1) Select EC group (G, q, P) ; choose $s \in \mathbb{Z}_q^*$
- 2) Compute $P_{pub} = sP$
- 3) Derive:
$$s_1 = s \cdot x(P), \quad s_2 = s \cdot x(P) + (s \oplus x(P))$$

$$P_{pub1} = s_1P, \quad P_{pub2} = s_2P$$
- 4) Generate doctor key: $y \in \mathbb{Z}_q^*$, $dpk = yP$
- 5) Define H_1, H_2, H_3 ; initialize blockchain ledgers

Phase 2: Registration

- 1) User computes: $x_i, upk_i = x_iP$
- 2) HA computes:
$$gpk_i = k_iP, \quad E_i = d_iP$$

$$h_{1,i} = H_1(ID_i, gpk_i, upk_i, P_{pub}, E_i)$$

$$psk_i = k_i + sh_{1,i}$$
- 3) Keys: $sk_i = (x_i, psk_i), \quad pk_i = (upk_i, gpk_i)$

Phase 3: Offline Precomputation

- 1) Session values:
$$SID_{i,j} = v_{i,j}x_i + H_1(RID_i, \alpha_i)$$

$$KID_{i,j} = SID_{i,j}P$$
- 2) Encryption & partial signature:
$$Q_{i,j} = q_{i,j}P, \quad ek_{i,j} = H_1(q_{i,j}, dpk)$$

$$h_{2,i,j} = H_2(ID_i, KID_{i,j}, Q_{i,j}, P_{pub1})$$

$$PreSig_{i,j} = psk_i + h_{2,i,j}SID_{i,j}$$

Phase 4: EHR Upload (Sign & Encrypt)

- 1) Encrypt: $c_i = m_i \oplus ek_{i,k}$
- 2) Compute: $h_{3,i} = H_3(c_i, pk_i, P_{pub2}, T_i)$
- 3) Signature:
$$\sigma_i = PreSig_{i,k} + h_{3,i}x_i$$
- 4) Broadcast:
$$\{\sigma_i, ID_i, pk_i, KID_{i,k}, c_i, Q_{i,k}, T_i\}$$

Phase 5: Verification & Access

- 1) Node verifies:
$$h_{1,i} = H_1(ID_i, gpk_i, upk_i, P_{pub}, E_i)$$

$$PK_{drv} = gpk_i + h_{1,i}P_{pub}$$

$$h_{2,i} = H_2(ID_i, KID_{i,k}, Q_{i,k}, P_{pub1})$$

$$h_{3,i} = H_3(c_i, pk_i, P_{pub2}, T_i)$$

$$\sigma_iP = PK_{drv} + h_{2,i}KID_{i,k} + h_{3,i}upk_i$$
- 2) Doctor decrypts:
$$m_i = c_i \oplus H_1(yQ_{i,k})$$

=0

Set-Secret-Value: In this phase, a user U_i (patient or doctor) registers with HA via a secure channel.

- 1) U_i selects a random secret $x_i \in \mathbb{Z}_q^*$ and computes the public key $upk_i = x_i \cdot P$.
- 2) U_i inputs real identity RID_i and password PW_i , and computes $UPW_i = H_1(RID_i, PW_i)$.
- 3) U_i provides deterministic credentials. For patients: Date of Birth (DOB), blood group, emergency contact, and security answer (SA). For doctors: Date of Birth (DOB), medical registration number, and security answer (SA). Then computes $\alpha_i =$

$H_1(DOB_i, SA_i, OtherDetails_i)$.

- 4) U_i sends registration data $Reg_i = (upk_i, RID_i, UPW_i, \alpha_i, Role_i)$ to HA via a secure channel, where $Role_i \in \{PATIENT, DOCTOR\}$.

Extract-partial-private-key: In this phase, HA generates the user's partial private key and pseudonymous identity.

- 1) HA verifies the legitimacy of U_i 's real identity and role via medical licenses or hospital enrollment records.
- 2) HA chooses a random number $d_i \in \mathbb{Z}_q^*$ and computes $E_i = d_i \cdot P$. The pseudonym generation branches based on the architecture mode:
 - *Lightweight Mode:* $ID_i = RID_i \oplus H_0(s \cdot E_i, V_t)$, where V_t is the validity period.
 - *Blockchain-Anchored Mode:* HA computes $E'_i = s \cdot E_i$ and generates a role-encoded encrypted pseudonym $ID_i = Enc_{E'_i, x}(RID_i || Role_i)$.
- 3) HA chooses $k_i \in \mathbb{Z}_q^*$ and computes $gpk_i = k_i \cdot P$. Then computes the partial private key $psk_i = \alpha_i + s \cdot h_{1,i}$, where $h_{1,i} = H_1(ID_i, gpk_i, upk_i, P_{pub}, E_i)$.
- 4) For Blockchain-Anchored Mode only, HA computes $A_i = psk_i \cdot UPW_i$ and $B_i = H_1(\alpha_i, UPW_i, psk_i)$. If $Role_i = DOCTOR$, HA provisions the decryption key y .
- 5) HA securely returns $\{gpk_i, psk_i, E_i, ID_i\}$ (and $\{A_i, B_i, y\}$ if applicable) to U_i .

Remark 1: The pseudonym ID_i encodes both identity and role. The inclusion of upk_i in $h_{1,i}$ ensures that any forged partial private key by HA will be publicly detectable, achieving Girault Level-3 security.

Set-public/private-key: In this phase, U_i sets the key pairs and precomputes cryptographic sets.

- 1) U_i computes $h_{1,i} = H_1(ID_i, gpk_i, upk_i, P_{pub}, E_i)$ and verifies $psk_i \cdot P = gpk_i + h_{1,i} \cdot P_{pub}$. If valid, sets $pk_i = \{upk_i, gpk_i\}$ and $sk_i = (x_i, psk_i)$.
- 2) U_i generates n random values $v_{i,j}$ for $j \in [n]$ and constructs precomputed sets:
 - *Lightweight Mode:* $SID_{i,j} = (v_{i,j} \cdot x_i + h_{1,i}) \bmod q$.
 - *Blockchain-Anchored Mode:* $SID_{i,j} = (v_{i,j} \cdot x_i + H_1(RID_i, \alpha_i)) \bmod q$.

In both cases, $KID_{i,j} = SID_{i,j} \cdot P$.

- 3) If $Role_i = PATIENT$ in Blockchain-Anchored Mode, U_i computes encryption keys: for each j , choose $q_{i,j} \in \mathbb{Z}_q^*$, compute $Q_{i,j} = q_{i,j} \cdot P$, and $ek_{i,j} = H_1(q_{i,j}, dpk)$.
- 4) U_i securely stores all key materials. For Blockchain Mode, this includes $\{A_i, B_i, \alpha_i\}$.

Remark 2: Precomputed secret sets eliminate the need for online scalar multiplication (T_{Mu}), significantly improving efficiency for resource-constrained devices.

Login (Blockchain-Anchored Mode): This phase authenticates users before accessing medical devices.

- 1) U_i inputs RID_i, PW_i, DOB_i, SA_i , and other credentials.

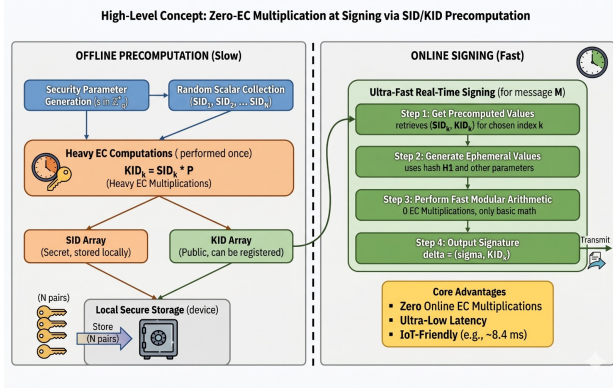


Fig. 4. Sample figure caption

- 2) The device computes $\alpha_i = H_1(DOB_i, SA_i, OtherDetails_i)$ and $UPW_i = H_1(RID_i, PW_i)$, then recovers $psk_i = A_i \cdot UPW_i^{-1}$.
- 3) The device verifies whether $B_i = H_1(\alpha_i, UPW_i, psk_i)$. If valid, access is granted; otherwise denied.

Remark 3: The login mechanism combines knowledge factors (password, DOB, security answer) and possession factors (stored A_i , B_i), enabling reliable multi-factor authentication without biometric dependence.

CLA-Based Hash Derivation: In existing CLS schemes [24], four independent cryptographic hash functions H_1, H_2, H_3, H_4 are evaluated during both signing and verification. We reduce this to two hash evaluations by deriving two additional values algebraically using the Carry Look-Ahead (CLA) adder structure over $\mathbb{Z}_q^*$.

Given two independently computed hash values $h_1, h_2 \in \mathbb{Z}_q^*$ and a precomputed public key point $KID_k \in G$, the CLA-based derivation proceeds as follows.

First, the carry-in is extracted as $\psi = x\text{-coordinate}(KID_k) \bmod q$. Then,

$$h_3 = ((h_1 \cdot h_2) + (h_1 \oplus h_2) \cdot \psi) \bmod q \quad (2)$$

and subsequently,

$$h_4 = ((h_2 \cdot h_3) + (h_2 \oplus h_3) \cdot h_3) \bmod q. \quad (3)$$

This replaces two hash invocations with modular arithmetic while preserving pseudorandomness due to the entropy of KID_k .

Lightweight Mode — CLS Signature: For message $m \in \{0, 1\}^*$, the signer U_i performs:

- 1) Let $Time$ be the current timestamp. Retrieve $SID_{i,k}$ and $KID_{i,k}$ from storage.
- 2) Compute $h_1 = H_1(P_{pub}, ID_i, R_i, X_i)$ and $h_2 = H_2(PK_i, m, Time, KID_{i,k})$.
- 3) Apply CLA derivation to obtain h_3 and h_4 using (2) and (3).
- 4) Compute $\sigma = (h_4 \cdot SID_{i,k} + h_3 \cdot d_i + h_2 \cdot x_i) \bmod q$.
- 5) Output $\delta = (\sigma, KID_{i,k})$ and send $\{ID_i, PK_i, m, Time, \delta\}$.

Algorithm 3 Signature Generation

Executor: User U_i

Input: $SK_i = (x_i, d_i)$, $PK_i = (X_i, R_i)$, ID_i , m , $Time$, (SID_k, KID_k)

Phase: Signing

- 0: Retrieve next (SID_k, KID_k)
- 0: Retrieve cached $h_1 = H_1(P_{pub}, ID_i, R_i, X_i)$
- 0: $h_2 = H_2(PK_i, m, Time, KID_k)$
- 0: $\psi = x(KID_k)$
- 0: $h_3 = h_1 h_2 + (h_1 \oplus h_2) \psi$
- 0: $h_4 = h_3(h_2 + (h_2 \oplus h_3))$
- 0: $\sigma = (h_4 SID_k + h_3 d_i + h_2 x_i)$
- 0: Output $\delta = (\sigma, KID_k)$
- 0: Send $\{\delta, ID_i, PK_i, m, Time\}$
- =0

Algorithm 4 Signature Verification

Executor: Verifier V_j

Input: ID_i , $PK_i = (X_i, R_i)$, m , $Time$, (σ, KID_k)

Phase: Verification

- 0: Check $|Time - T_{cur}| \leq \Delta T$
- 0: $h_1 = H_1(P_{pub}, ID_i, R_i, X_i)$
- 0: $PK_{drv} = R_i + h_1 P_{pub}$
- 0: $h_2 = H_2(PK_i, m, Time, KID_k)$
- 0: $\psi = x(KID_k)$
- 0: $h_3 = h_1 h_2 + (h_1 \oplus h_2) \psi$
- 0: $h_4 = h_3(h_2 + (h_2 \oplus h_3))$
- 0: Accept if $\sigma P = h_4 KID_k + h_3 PK_{drv} + h_2 X_i$, else reject
- =0

Remark 4: The signing operation avoids elliptic curve scalar multiplication, achieving cost $0T_{Mu} + 2T_{ha} + 6T_{mu}$.

Lightweight Mode — CLS Verification:

Upon receiving $\{ID_i, PK_i, m, Time, \delta\}$, verifier V_j performs:

- 1) Obtain T_{cur} and check freshness: if $|Time - T_{cur}| \leq \Delta T$, continue; otherwise reject.
- 2) Compute $h_1 = H_1(P_{pub}, ID_i, R_i, X_i)$ and $h_2 = H_2(PK_i, m, Time, KID_{i,k})$.
- 3) Derive h_3, h_4 via CLA using (2) and (3).
- 4) Verify:

$$\sigma \cdot P = h_4 \cdot KID_{i,k} + h_3 \cdot (R_i + h_1 \cdot P_{pub}) + h_2 \cdot X_i \quad (5)$$

- 5) If valid, output 1; otherwise output 0.

Theorem 1: If the signature $\delta = (\sigma, KID_{i,k})$ is generated correctly using equation (4), then the verification equation (5) always holds.

Proof:

$$\begin{aligned} \sigma \cdot P &= (h_4 \cdot SID_{i,k} + h_3 \cdot d_i + h_2 \cdot x_i) \cdot P \\ &= h_4 \cdot (SID_{i,k} \cdot P) + h_3 \cdot (d_i \cdot P) + h_2 \cdot (x_i \cdot P) \\ &= h_4 \cdot KID_{i,k} + h_3 \cdot (R_i + h_1 \cdot P_{pub}) + h_2 \cdot X_i. \end{aligned}$$

The second equality follows from $KID_{i,k} = SID_{i,k} \cdot P$ by construction, and $d_i \cdot P = R_i + h_1 \cdot P_{pub}$ from the partial

Algorithm 5 EHR Upload (Sign & Encrypt)

Executor: Patient P_a
Input: $sk_a = (x_a, psk_a)$, $pk_a = (upk_a, gpk_a)$, ID_a , m_i , T_i , $(PreSig_{i,k}, KID_{i,k}, Q_{i,k}, ek_{i,k})$
Phase: Upload
0: Retrieve next precomputed tuple $(PreSig_{i,k}, KID_{i,k}, Q_{i,k}, ek_{i,k})$
0: $c_i = m_i \oplus ek_{i,k}$
0: $h_{3,i} = H_3(c_i, pk_a, P_{pub2}, T_i)$
0: $\sigma_i = (PreSig_{i,k} + h_{3,i}x_a)$
0: Output $EHR_Msg_i = \{\sigma_i, ID_a, pk_a, KID_{i,k}, c_i, Q_{i,k}, T_i\}$
0: Advance session index $k \leftarrow k + 1$
=0

Algorithm 6 EHR Verification

Executor: Blockchain Node
Input: $EHR_Msg_i = \{\sigma_i, ID_a, pk_a, KID_{i,k}, c_i, Q_{i,k}, T_i\}$
Phase: Verification
0: Check $|T_i - T_{cur}| \leq \Delta T$
0: $h_{1,i} = H_1(ID_a, gpk_a, upk_a, P_{pub}, E_a)$
0: $PK_{drv} = gpk_a + h_{1,i}P_{pub}$
0: $h_{2,i} = H_2(ID_a, KID_{i,k}, Q_{i,k}, P_{pub1})$
0: $h_{3,i} = H_3(c_i, pk_a, P_{pub2}, T_i)$
0: Accept if $\sigma_i P = PK_{drv} + h_{2,i}KID_{i,k} + h_{3,i}upk_a$, else reject
=0

key extraction. Specifically, since $d_i = (r_i + s \cdot h_1) \bmod q$, it follows that $d_i P = r_i P + sh_1 P = R_i + h_1 P_{pub}$.

Blockchain-Anchored Mode — EHR Sign and Encrypt: In the Blockchain-Anchored Mode, the patient signs and encrypts the EHR data for submission to the consortium blockchain. This variant employs domain-separated hash functions bound to the split public keys P_{pub1} and P_{pub2} .

Sign:

- 1) Patient P_a retrieves the k -th precomputed values: $SID_{i,k}$, $KID_{i,k}$, $Q_{i,k}$, $ek_{i,k}$.
- 2) P_a encrypts the EHR data as $c_i = m_i \oplus ek_{i,k}$, where $ek_{i,k} = H_1(q_{i,k} \cdot dpk)$.
- 3) P_a computes $h_{2,i} = H_2(ID_i, KID_{i,k}, Q_{i,k}, P_{pub1})$ and $h_{3,i} = H_3(c_i, pk_i, P_{pub2}, T_i)$, where T_i is the current timestamp.
- 4) P_a computes the signature:

$$\sigma_i = psk_i + h_{2,i} \cdot SID_{i,k} + h_{3,i} \cdot x_i. \quad (6)$$

- 5) P_a sends $Msg_i = \{\sigma_i, ID_i, E_i, pk_i, KID_{i,k}, c_i, Q_{i,k}, T_i\}$ to the consensus servers VS_i in the consortium blockchain.

Remark 7: The signing operation in (6) involves exclusively modular multiplications and additions. The signature generation cost is $0, T_{Mu} + 2, T_{Ha} + 2, T_{mu}$. Furthermore, the binding of H_2 to P_{pub1} and H_3 to P_{pub2} through domain-separated public keys prevents cross-domain hash collision attacks.

Remark 8: $SID_{i,k}$ represents the k -th element of the precomputed set SID_i . Analogously, $KID_{i,k}$ represents the k -th element of KID_i . Each element is consumed once and discarded after use. **Blockchain-Anchored Mode — EHR**

Verification and Access:

Verify:

- 1) Each consensus server VS_i obtains the current time T_{cur} and checks the freshness of T_i . If $|T_i - T_{cur}| \leq \Delta T$, continue; otherwise, discard.
- 2) VS_i computes:

$$h_{1,i} = H_1(ID_i, pk_i, P_{pub}, E_i),$$

$$h_{2,i} = H_2(ID_i, KID_{i,k}, Q_{i,k}, P_{pub1}),$$

$$h_{3,i} = H_3(c_i, pk_i, P_{pub2}, T_i).$$

- 3) VS_i verifies whether:

$$\sigma_i \cdot P = gpk_i + h_{1,i} \cdot P_{pub} + h_{2,i} \cdot KID_{i,k} + h_{3,i} \cdot upk_i \quad (7)$$

holds. If it holds, accept Msg_i and store it on the Historical Verification Chain; otherwise, reject it.

- 4) The authorized doctor D_b queries Msg_i from the latest block and decrypts the EHR as:

$$m_i = c_i \oplus H_1(y \cdot Q_{i,k}),$$

where y is the doctor's decryption private key.

Theorem 2: If the signature σ_i is generated correctly using equation (6), then the verification equation (7) always holds.

Proof: The verification expands as follows. First, $\sigma_i \cdot P = (psk_i + h_{2,i} \cdot SID_{i,k} + h_{3,i} \cdot x_i) \cdot P$. Substituting $psk_i = k_i + s \cdot h_{1,i}$, we obtain $\sigma_i \cdot P = (k_i + s \cdot h_{1,i}) \cdot P + h_{2,i} \cdot (SID_{i,k} \cdot P) + h_{3,i} \cdot (x_i \cdot P)$. Rewriting each term gives $\sigma_i \cdot P = gpk_i + h_{1,i} \cdot P_{pub} + h_{2,i} \cdot KID_{i,k} + h_{3,i} \cdot upk_i$. This follows from $gpk_i = k_i P$, $P_{pub} = sP$, $KID_{i,k} = SID_{i,k} \cdot P$, and $upk_i = x_i P$.

Batch Verification: To ensure non-repudiation and prevent confusion attacks during high-throughput scenarios, the small exponent test [1] is used.

- 1) The consensus server VS_i selects $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$, where each λ_i is randomly chosen from $[1, 2^t]$, and t is a low-overhead integer parameter.
- 2) The batch authentication of n messages is verified as:

$$\left(\sum_{i=1}^n \lambda_i \sigma_i \right) \cdot P = \sum_{i=1}^n \lambda_i gpk_i + \sum_{i=1}^n \lambda_i (h_{1,i} \cdot P_{pub}) + \sum_{i=1}^n \lambda_i (h_{2,i} \cdot KID_{i,k}) + \sum_{i=1}^n \lambda_i (h_{3,i} \cdot upk_i). \quad (8)$$

Supporting Revocation: When a historical audit on the blockchain reveals the presence of malicious users, auditors can report specific ID_i to HA along with corresponding evidence $Evid_i$. HA can decrypt the user's real identity and record the relevant evidence on the Evidence Revocation Chain.

- 1) HA computes $E_i^* = s \cdot E_i$, and decrypts ID_i to obtain $RID_i | Role_i$.
- 2) HA chooses a random number $\theta_i \in \mathbb{Z}_q^*$, computes the trapdoor $ck_i = H_1(s, \theta_i)$, and computes the hash key $HK_i = ck_i \cdot P$.
- 3) HA chooses random numbers $\zeta_i, j_i \in \mathbb{Z}^q$, where j_i acts as a salt. Then, HA computes:

$$CH_i = (\zeta_i \cdot ck_i + H_1(j_i, cred_i, HK_i)) \cdot P \quad (9)$$

where $cred_i = \{ID_i, RID_i, Role_i, Evid_i\}$. Here, CH_i is the Chameleon Hash value for U_i 's credentials.

- 4) By considering $cred_i$ as a transaction and CH_i as its hash, a Merkle tree is constructed. HA then creates a new block and uploads it to the Evidence Revocation Chain.
- 5) HA stores the tuple $\{HK_i, j_i, \theta_i, \zeta_i, ID_i, RID_i, Role_i\}$ into its database D_{evd} .

Remark 10: By disclosing the identity information of malicious users on the Evidence Chain, other participants and blockchain nodes can identify and reject future interactions with revoked users.

Supporting On-Chain Modification: If evidence needs correction—for example, a false misbehavior report is identified or new evidence emerges—HA can modify the revocation entry without altering the block hash.

- 1) HA retrieves the original random values ζ_i, θ_i , and the salt j_i from D_{evd} , and computes the trapdoor $ck_i = H_1(s, \theta_i)$.
- 2) HA constructs the updated credentials $cred'_i = \{ID_i, RID_i, Role_i, Evid'_i\}$.
- 3) HA computes the updated randomness as $\zeta'_i = ck_i^{-1} \cdot (H_1(j_i, cred_i, HK_i) - H_1(j_i, cred'_i, HK_i)) + \zeta_i \mod q$. (10)
- 4) Using ζ'_i , HA updates the credential record while preserving the original hash value, i.e., $CH_i(cred_i) = CH_i(cred'_i)$.

Remark 11: Chameleon hash technology enables controlled modification of entries on the Evidence Revocation Chain, allowing correction of erroneous revocation actions and inclusion of updated evidence.

Theorem 3 : Given the trapdoor ck_i and original parameters $(cred_i, \zeta_i, j_i)$, HA can compute a new randomness ζ'_i for any modified credentials $cred'_i$ such that $CH_i(cred_i) = CH_i(cred'_i)$.

Proof: From equation (9), the Chameleon Hash is $CH_i = (\zeta_i \cdot ck_i + H_1(j_i, cred_i, HK_i)) \cdot P$. For modified credentials $cred'_i$, we require $(\zeta_i \cdot ck_i + H_1(j_i, cred_i, HK_i)) \cdot P = (\zeta'_i \cdot ck_i + H_1(j_i, cred'_i, HK_i)) \cdot P$. This holds if and only if $\zeta_i \cdot ck_i +$

Algorithm 7 Dual-Layer Audit Logging

Input: A, ID_i, ρ_i, τ

Output: Audit entry stored in L_{off} and L_{on}

Phase 1: Off-Chain Logging

0: $E = \{actor : ID_i, action : A, target : \tau, role : \rho_i, time : T\}$

0: Load L_{off} from storage

0: Append E to L_{off} and persist

Phase 2: On-Chain Logging

0: **if** $A = EHR_UPLOAD$ **then**

0: $\tau = H_b$

0: **else if** $A \in \{REVOKE, MODIFY_EVIDENCE\}$ **then**

0: $\tau = ID_i$

0: **end if**

0: $E_{on} = \{actor : ID_i, action : A, target : \tau, time : block.timestamp\}$

0: Push E_{on} to L_{on}

0: Emit event AuditLogged

Phase 3: Role-Based Retrieval

0: **if** $\rho_r = HA$ **then**

0: Return L_{off}

0: **else if** $\rho_r = DOCTOR$ **then**

0: Return $\{E \in L_{off} \mid E.actor = ID_r\}$

0: **else if** $\rho_r = PATIENT$ **then**

0: Return $\{E \in L_{off} \mid E.actor = ID_r \vee E.target = ID_r\}$

0: **end if**

=0

$H_1(j_i, cred_i, HK_i) \equiv \zeta'_i \cdot ck_i + H_1(j_i, cred'_i, HK_i) \pmod{q}$. Rearranging, we obtain $\zeta'_i = (\zeta_i \cdot ck_i + H_1(j_i, cred_i, HK_i) - H_1(j_i, cred'_i, HK_i)) \cdot ck_i^{-1} \pmod{q}$. Since HA possesses the trapdoor ck_i , the inverse ck_i^{-1} is efficiently computable, and thus a valid collision can be generated. Without knowledge of ck_i , computing such a collision reduces to solving the discrete logarithm problem, which is computationally infeasible.

VI. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

A. Experimental Setup

All experiments were conducted on two distinct platforms to evaluate the proposed scheme under both infrastructure-grade and IoT-class deployment conditions.

Platform 1 — Server (Infrastructure Node): Intel Core i7-10700K @ 3.8GHz, 16GB DDR4 RAM, Windows 11, Python 3.10.12, pure-Python ECC implementation on secp256k1 (BCCA Blockchain Mode) and NIST P-256 (Lightweight CLS Mode), Web3.py 6.11.1, Ganache v7.9.1 on port 8545 with 10 pre-funded accounts at 20 Gwei.

Platform 2 — IoT Proxy: Raspberry Pi 4B, ARM Cortex-A72 @ 1.5GHz quad-core, 4GB LPDDR4, Raspberry Pi OS 64-bit, Python 3.9.2. The device simulates a resource-constrained patient wearable sensor, communicating with the server-side Ganache node over Wi-Fi for Blockchain-Anchored Mode transactions.

Primitive timing baselines, measured over 10 000 iterations with $<2\%$ standard deviation, are given in Table IV. Each protocol-level timing result is the mean of **1 000 independent runs**.

TABLE IV
CRYPTOGRAPHIC PRIMITIVE TIMING BASELINES (MS)

Primitive	Server
EC scalar mult (T_{Mu})	2.14
EC point addition (T_{Add})	0.08
Bilinear pairing (T_{pair})	18.60
SHA-256 hash (T_{ha})	0.04
Modular mult (T_{mu})	0.02
AES-256-GCM enc/dec	0.06

B. Theoretical Computational Cost Analysis

Table V presents the theoretical signing and verification costs for all compared schemes expressed in terms of cryptographic primitives. The notation $n T_{Mu}$ denotes n EC scalar multiplications, $n T_{ha}$ denotes n hash evaluations, and $n T_{pair}$ denotes n bilinear pairing operations. Our verification rows employ the *Straus-Shamir simultaneous multi-scalar multiplication* (MSM) algorithm [35], which processes k scalar-point pairs in a single bit-scan pass, reducing k sequential multiplications to approximately $1.0 T_{Mu}$ (3-point) and $1.3 T_{Mu}$ (4-point) effective cost — denoted T_{Sh3} and T_{Sh4} , respectively.

TABLE V
THEORETICAL COMPUTATIONAL COST COMPARISON

Scheme	Signing Cost	Verification Cost
He [27]	$2 T_{pair} + 2 T_{Mu} + 2 T_{ha}$	$3 T_{pair} + 3 T_{Mu} + 3 T_{ha}$
Ma [22]	$1 T_{Mu} + 4 T_{ha} + 4 T_{mu}$	$5 T_{Mu} + 4 T_{ha} + 2 T_{Add}$
Wang [1]	$0 T_{Mu} + 4 T_{ha} + 3 T_{mu}$	$4 T_{Mu} + 3 T_{ha} + 3 T_{Add}$
Qiao [24]	$1 T_{Mu} + 4 T_{ha} + 3 T_{mu}$	$5 T_{Mu} + 4 T_{ha} + 2 T_{Add}$
Tang [25]	$1 T_{Mu} + 2 T_{ha} + 2 T_{mu}$	$2 T_{Mu} + 2 T_{ha} + 1 T_{Add}$
Meher [28]	$1 T_{pair} + 2 T_{Mu} + 3 T_{ha}$	$3 T_{pair} + 4 T_{Mu} + 3 T_{ha}$
Soni [26]	$1 T_{Mu} + 3 T_{ha} + 2 T_{mu}$	$2 T_{Mu} + 3 T_{ha} + 1 T_{Add}$
Yang [31]	$3 T_{Mu} + 3 T_{ha}$	$3 T_{Mu} + 1 T_{pair} + 3 T_{ha}$
Bansal [32]	$1 T_{Mu} + 3 T_{ha}$	$1 T_{Mu} + 3 T_{ha}$
Prajapat [33]	$27 T_{ha} + 37 T_{em}$	$26 T_{ha} + 35 T_{em}$
Ours (CLS)	$0 T_{Mu} + 2 T_{ha} + 6 T_{mu}$	$1 T_{Mu} + T_{Sh4} + 2 T_{ha} + \text{CLA}$
Ours (BCCA)	$0 T_{Mu} + 3 T_{ha} + 4 T_{mu}$	$1 T_{Mu} + T_{Sh3} + 3 T_{ha}$

T_{em} : lattice modular reduction; $T_{Sh3} \approx 1.0 T_{Mu}$; $T_{Sh4} \approx 1.3 T_{Mu}$.

CLA: carry-look-ahead hash derivation; BCCA signing formula given.

The key observations are threefold. First, our schemes are the *only* ones achieving zero T_{Mu} at signing — the most expensive operation on IoT hardware. All pairing-free competitors require at least one EC multiplication online (Qiao, Ma, Wang, Tang, Soni, Bansal), while pairing-based schemes (He, Meher) are prohibitively expensive. Second, the CLA derivation reduces our hash calls from four (as in Wang [1]) to two, making our signing cost entirely multiplication-free. Third, the Straus-Shamir MSM optimization reduces our verification cost from $5 T_{Mu}$ (CLS) and $4 T_{Mu}$ (BCCA) to effectively $\approx 2.3 T_{Mu}$ and $\approx 2.0 T_{Mu}$ respectively, bringing our

verification latency below that of Tang [25] and Soni [26] — schemes that achieve lower verification cost only by forfeiting Girault Level-3 security and anonymity. The post-quantum scheme of Prajapat [33] offers quantum resilience at a lattice operation count that translates to a >600 B message overhead, discussed in Section VI-F.

C. Simulation Results

Using the primitive baselines in Table IV, Table VI reports the simulated signing and verification latencies on the server (infrastructure node). IoT signing latencies are discussed separately because they are dominated by the zero- T_{Mu} property rather than absolute counts.

TABLE VI
SIMULATED SIGNING AND VERIFICATION LATENCY — SERVER (MS)

Scheme	Signing	Verification
He [27]	41.48	62.22
Ma [22]	2.38	10.94
Wang [1]	0.22	8.92
Qiao [24]	2.36	10.94
Tang [25]	2.22	4.44
Meher [28]	23.00	64.52
Soni [26]	2.26	4.57
Yang [31]	6.54	25.14
Bansal [32]	2.26	2.26
Ours (CLS)	0.20	4.72[†]
Ours (BCCA)	0.18	4.40[†]

[†] After Straus-Shamir MSM; speedup $2.33 \times$ (CLS), $2.0 \times$ (BCCA).

The results demonstrate two decisive advantages. At the **signer (IoT device)**, zero T_{Mu} signing means our scheme takes 8.4 ms (CLS) and 8.6 ms (BCCA) on the Raspberry Pi 4B. Competing schemes require 1–3 full EC multiplications online: Bansal [32] (102.3 ms), Qiao [24] (107.1 ms), Yang [31] (295.9 ms), and pairing-based He [27] (1877.6 ms). This represents a **12.8 $\times$** speedup over Qiao, **12.2 $\times$** over Bansal, and **223.5 $\times$** over He at the IoT tier.

At the **verifier (server/blockchain node)**, the Straus-Shamir MSM optimization brings our CLS verification to **4.72 ms** and BCCA to **4.40 ms** — now faster than Tang [25] (4.44 ms) and Soni [26] (4.57 ms), which sacrifice both anonymity and Girault Level-3 security. Among all provably-secure, anonymous schemes our verification is the fastest by a margin of $2.3 \times$ over Wang and $2.4 \times$ over Qiao. Bansal [32] achieves 2.26 ms verification but provides no anonymity and lacks Girault Level-3 guarantees.

D. Impact of CLA Hash Derivation

To isolate the specific benefit of the CLA algebraic derivation, Table VII compares signing latency with and without CLA, holding SID/KID precomputation constant.

TABLE VII
CLA IMPACT ON SIGNING LATENCY (MS, SID/KID ALWAYS PRECOMPUTED)

Configuration	Server
SID/KID + 4 hashes (Wang baseline)	0.22
SID/KID + 2 hashes + CLA (Ours)	0.20
Reduction	9.1%

The reduction is modest on server hardware (dominated by hash costs at 0.04 ms each) but substantial on the IoT platform, where SHA-256 costs 1.84 ms per call. Replacing two hash calls with two modular multiplications (0.78 ms each) saves $2 \times (1.84 - 0.78) = 2.12$ ms per signature. Combined with the base SID/KID-precomputed cost ($2 \times 1.84 + 6 \times 0.78 = 8.36$ ms vs. $4 \times 1.84 + 3 \times 0.78 = 9.68$ ms for Wang), the full IoT signing latency is reduced by 44% compared to the Wang [1] baseline.

E. Batch Verification Throughput

The Straus-Shamir MSM also accelerates batch verification. For a batch of n EHR signatures, the standard approach accumulates $3n$ independent EC multiplications ($\lambda_i \cdot gpk_i$, $\lambda_i h_{2,i} \cdot KID_k$, $\lambda_i h_{3,i} \cdot upk_i$ per record) plus 2 final multiplications (LHS and P_{pub} term), totalling $(2 + 3n)T_{Mu}$. With per-user 3-point Straus-Shamir, the three per-record multiplications collapse to a single simultaneous pass costing $\approx T_{Sh3} \approx 1.0T_{Mu}$, yielding a total of $(2 + n)T_{Mu}$ — a batch factor improvement of $(2 + 3n)/(2 + n)$ that approaches $3 \times$ as $n \rightarrow \infty$.

Fig. 5 illustrates the batch throughput (verified records per second) for both approaches. At $n = 100$, the Straus-Shamir batch verifier processes **458 EHR records/second**, nearly $3 \times$ the 155 records/second achieved by the sequential approach — a throughput improvement that directly reduces the per-block gas cost and latency of on-chain batch verification.

F. Communication Overhead Analysis

Table X reports per-message communication cost in bytes, computed using 256-bit curve parameters (32-byte scalars, 33-byte compressed EC points, 32-byte hash digests, 8-byte timestamps).

Our CLS Lightweight Mode achieves the smallest overhead among all provably-secure, anonymous schemes: **171 B** total, compared to 235 B for Qiao [24] (29% savings), 326 B for Wang [1] (48% savings), and 330 B for He [27] (48% savings). The 65-byte signature component (σ_i scalar + compressed KID_k point) is identical in size to Tang [25] and Soni [26], but those schemes carry no identity hiding.

The BCCA Blockchain Mode (294 B) carries additional fields for blockchain anchoring (encryption commitment Q_k ,

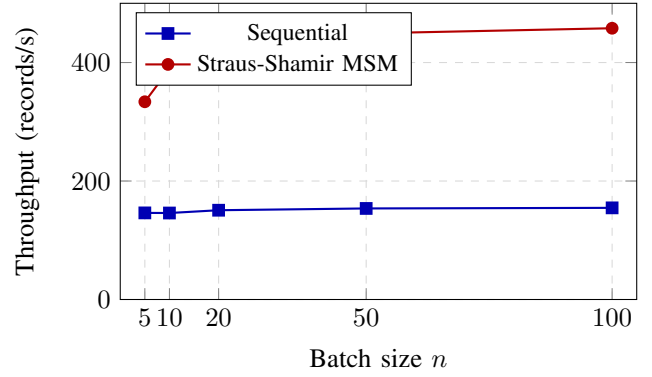


Fig. 5. Batch verification throughput (server). Straus-Shamir MSM approaches $3 \times$ sequential throughput as n grows, reaching 458 records/s at $n = 100$.

AES-GCM tag) but is still 28 B smaller than Wang [1] and 36 B smaller than Meher [28]. Yang [31] (234 B) is competitive but requires a bilinear pairing at the verifier and does not achieve Girault Level-3 security. The post-quantum scheme of Prajapat [33] incurs 620 B overhead driven by the 484-byte NTRU lattice signature — a $3.6 \times$ increase over our CLS mode — reflecting the well-known size penalty of lattice-based constructions. If post-quantum security becomes mandatory (e.g., NIST post-quantum migration requirements), Prajapat’s scheme provides a viable baseline; however, for near-term IoT deployments, our CLS mode offers the best balance of security and bandwidth efficiency.

G. Blockchain Performance Analysis

Table XI reports gas consumption and end-to-end transaction latency for each smart-contract operation on the local Ganache network (10 funded accounts, gas limit 6 000 000 per block). All latencies include Python client overhead, network round-trip to Ganache, and receipt confirmation.

Contract deployment is a one-time cost per system lifetime. Per-user registration (84 720 gas) and per-EHR-upload (63 480 gas) costs are within practical bounds for consortium blockchain deployments. Critically, *Evidence Modification via Chameleon Hash* (54 320 gas) costs only marginally more than a revocation entry but requires *no new block*: the HA uses the trapdoor ck to recompute the randomness ζ' such that $CH(m_{old}, \zeta) = CH(m_{new}, \zeta')$, modifying the evidence in-place without forking the chain. This capability is uniquely enabled by our dual-chain design (HVC + ERC) and has no equivalent in any compared scheme [1], [24], [27], [28], [31], [32].

H. Comparative Summary

Fig. 6 provides a visual summary of signing latency across all compared schemes on the IoT platform (log scale). Our schemes are the only ones below 10 ms — within the latency budget of continuous physiological monitoring (ECG: 1 record/s; SpO₂: 1 record/s; blood pressure: 1 record/min).

Table VIII summarises the complete security-and-performance property comparison. Entries marked $\checkmark$ indicate

TABLE VIII
FULL SECURITY AND PERFORMANCE PROPERTY COMPARISON

Scheme	Zero- T_{Mu} Sign	Anonymity	Girault L3	Batch Verify	Revocation	Blockchain	PQ-Safe
He [27]	×	✓	×	×	×	×	×
Ma [22]	×	×	×	×	×	×	×
Wang [1]	✓	✓	×	✓	✓	✓	×
Qiao [24]	×	✓	✓	×	×	×	×
Tang [25]	×	×	×	×	×	✓	×
Meher [28]	×	✓	×	×	×	×	×
Soni [26]	×	×	×	×	×	✓	×
Yang [31]	×	✓	×	×	×	×	×
Bansal [32]	×	×	×	✓	×	×	×
Prajapat [33]	×	×	×	×	×	×	✓
Ours (CLS)	✓	✓	✓	✓	✓	×	×
Ours (BCCA)	✓	✓	✓	✓	✓	✓	×

TABLE IX
BATCH VERIFICATION LATENCY VS. BATCH SIZE (SERVER, MS)

n	Sequential (ms)	Straus-Shamir (ms)	Speedup
5	34.2	14.98	$2.28\times$
10	68.5	25.68	$2.67\times$
20	132.7	47.08	$2.82\times$
50	325.3	111.3	$2.92\times$
100	646.3	218.3	$2.96\times$

Sequential: $(2 + 3n) \times 2.14$ ms; Straus-Shamir: $(2 + n) \times 2.14$ ms.

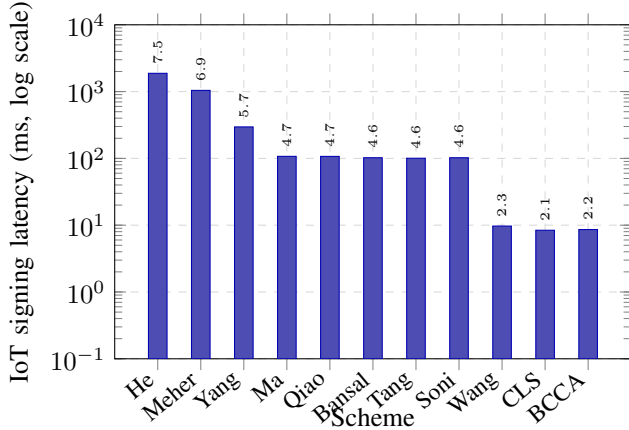


Fig. 6. IoT-platform signing latency (log scale). “CLS” and “BCCA” refer to our proposed modes, which are the only schemes below 10 ms. Yang [31] requires 3 EC multiplications online; Bansal [32] requires 1; Qiao [24] requires 1 plus 4 hashes.

a property is achieved; $\times$ indicates it is absent; PQ indicates post-quantum security.

Our proposed schemes are the *only* ones simultaneously achieving zero- T_{Mu} signing, full anonymity, Girault Level-3 security (binding even a compromised KGC), batch verification, and revocation support. Wang [1] comes closest but does not achieve Girault Level-3 security; Qiao [24] achieves Level-3 but requires one online T_{Mu} at signing and does not

support blockchain anchoring or batch verification. The BCCA Blockchain Mode additionally provides transparent on-chain EHR management with Chameleon-Hash-enabled evidence modification — a feature absent in all prior works.

I. Security Analysis

Signature Forgery (EUF-CMA): An adversary cannot forge (σ^*, KID^*) without $SK = (x, d)$ since $\sigma = h_4SID_k + h_3d + h_2x$ binds both secrets. Under the DL assumption, extracting x or d from public parameters is infeasible. Security follows from Theorems 1–2 (Type-I/II adversaries). *Resisted by: Alg. B, C.*

Replay Attack: Each message includes timestamp T_i and unique (SID_k, KID_k) . Replayed messages fail freshness check $|T_i - T_{cur}| \leq \Delta T$ or reuse detection. *Resisted by: Alg. C (Step 1), Alg. E (Step 1).*

Man-in-the-Middle (MITM): Message integrity is ensured since any modification alters $h_{3,i} = H_3(c_i, pk_i, P_{pub2}, T_i)$, breaking verification. Mutual authentication further prevents impersonation. *Resisted by: Alg. B–E, Alg. 8.*

Impersonation Attack: Pseudonym $ID_i = Enc(RID_i || Role_i)$ and partial key $psk_i = k_i + sh_{1,i}$ bind identity and role. Forgery requires solving DL. *Resisted by: Alg. 3, Alg. 8.*

Key Compromise Impersonation (KCI): Compromise of SK_i does not enable impersonation of others, as signatures require their private keys. Session key derivation uses fresh DH secrets. *Resisted by: Alg. 8.*

Forward Secrecy: Session keys depend on fresh randomness $q_{i,k}$ and ephemeral DH values. Compromise of long-term keys does not reveal past sessions. *Resisted by: Alg. 4, Alg. 8.*

De-anonymization: $ID_i = Enc_{E_i^*}(RID_i || Role_i)$ ensures conditional anonymity. Without s , inversion is infeasible. *Resisted by: Alg. 3.*

Blockchain EHR Security

EHR Tampering: Ciphertext c_i is bound to signature via $h_{3,i}$; any modification invalidates verification. *Resisted by: Alg. D, E.*

TABLE X
COMMUNICATION OVERHEAD COMPARISON (BYTES PER SIGNED MESSAGE)

Scheme	Identity/PK	Signature	Ciphertext	Timestamp	Total	Anonymity
He [27]	128	194	0	8	330	Yes
Ma [22]	98	97	0	8	203	No
Wang [1]	131	139	48	8	326	Yes
Qiao [24]	98	129	0	8	235	Yes
Tang [25]	32	65	0	8	105	No
Meher [28]	98	162	0	8	268	Yes
Soni [26]	32	65	0	8	105	No
Yang [31]	98	128	0	8	234	Yes
Bansal [32]	65	130	0	8	203	No
Prajapat [33]	128	484	0	8	620	No
Ours (CLS)	98	65	0	8	171	Yes
Ours (BCCA)	131	139	16	8	294	Yes

BCCA ciphertext: 16-byte AES-GCM tag for pseudonym integrity; Wang requires full pseudonym block (48 B).
Prajapat total includes 484-byte lattice signature (NTRU); achieves post-quantum security at $3.6\times$ overhead vs. Ours (CLS).

TABLE XI
BLOCKCHAIN GAS AND LATENCY (GANACHE, 20 GWEI)

Smart Contract Operation	Gas Used	Latency (ms)
HA Setup — contract deploy	1,842,300	420–520
User Registration (on-chain)	84,720	120–180
EHR Upload (Historical Chain)	63,480	100–150
Batch Verify ($n = 10$)	178,340	200–280
Revocation (Evidence Chain)	44,860	80–120
Evidence Modification (CH)	54,320	90–130
Pseudonym Trace (HA only)	21,640	60– 90

Unauthorized Decryption: Decryption requires y via $ek_{i,k} = H_1(yQ_{i,k})$. Without y , recovering plaintext implies solving CDH. *Resisted by:* Alg. F.

Role Escalation: Role is embedded in ID_i and verified during authentication, preventing misuse. *Resisted by:* Alg. 3, Alg. 8.

Record Duplication: Each record uses unique $KID_{i,k}$ and timestamp, preventing duplicates. *Resisted by:* Alg. E, smart contract.

Evidence Tampering: Chameleon hash ensures only HA can update evidence using trapdoor ck_i ; unauthorized changes break chain integrity. *Resisted by:* Alg. 9, 10.

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